How Much of the Learned Testing Policy Should a Model Keep?

Adapting deep clinical time-series models to observation shift from a small run-in

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1 Project idea and research question

Deep models for clinical time series learn from *what* is measured and also from *when* and *whether* it is measured: a lactate order signals that a doctor suspects sepsis. This testing pattern is predictive [1], but it reflects the hospital's observation policy, and it changes when a model trained at hospital A is deployed at hospital B (*observation shift*). The change comes in two kinds. Hospital B may test everyone more, so every order means a bit less; or individual tests may change meaning, for example troponin is ordered for 3% of patients at Beth Israel but 32% at Emory (PhysioNet 2019 [2]), while a lactate order may keep its meaning. A model adapted to B must fix the first kind globally and the second per test.

Before a model goes live, hospitals often run it silently for 1–3 months [3]. That run-in yields the only labelled data from B, and very little of it: a typical ICU admits 600–1,300 patients a year [4], so a 1–3 month run-in contains about 2–6 sepsis cases. We therefore propose a **ladder of adaptations** that changes only the testing part of a latent state-space model in which the testing evidence enters through its own head: (1) one global weight w on the testing evidence (w=0 ignore, w=1 trust as at A); (2) one weight per group of tests; (3) re-learning the testing head with the rest frozen; (4) fine-tuning the whole model. Higher rungs repair more kinds of shift but need more outcome cases.

Research question. *For a run-in of a given size at a new hospital, which rung of adaptation of the testing part predicts best, and how does the answer depend on the kind of observation shift?* Concretely: (i) does adapting only the testing part beat the fixed rules of prior work (always keep or always remove the pattern) and full fine-tuning at realistic run-in sizes; (ii) at how many outcome events does each higher rung start to pay off; (iii) does the state-space model adapt better from few cases than a standard architecture (GRU-D) fine-tuned at its last layer or in full.

2 Literature overview

Deep models use the observation pattern but do not separate it. Architectures for irregular series consume masks and time gaps (GRU-D [5]), continuous-time paths (Neural CDE [6], Latent ODE [7]) or time-value sets and attention (mTAN [8], Raindrop [9]), and fuse the pattern with physiology in one representation. Models with a separate observation component exist, a Hawkes process for clinicians' measurement events [10] and DeepJoint's observation heads [11], but none adapts that component alone at a new site.

Observation shift between hospitals. Multi-site deep sepsis and ICU benchmarks lose 0.08–0.09 AUROC across sites [12, 13] and include missingness indicators without attributing the loss to them. Studies that do isolate the pattern choose all or nothing: Futoma et al. [14] fitted with measurement indicators and dropped them at deployment, which transferred best from Beth Israel to a Memphis hospital but not in reverse; Rockenschaub et al. [15] found that ignoring informative missingness costs at the source but is more robust elsewhere, especially when missingness is highly informative. Lasko et al. [16] argue conceptually for a site-specific process model next to a stable disease model; Zhou et al. [17] adapt to missingness shift without labels in linear models. None tunes *how much* of the learned policy to keep from labelled target data.

Adapting with few labels. Clinical model updating is a ladder from recalibration to full refitting, and parsimonious updates win at small samples [18, 19]; a full refit is already risky well below the 100 events needed just to validate a model [20]. In deep learning, which part to fine-tune depends on the kind of shift [21], and retraining only the last layer can suffice [22], but clinical fine-tuning studies use hundreds to thousands of target patients [12, 13]. **Gap:** no work adapts only the observation part of a clinical sequence model, and none compares such adaptations against run-in size.

Preliminary results (linear models, open PhysioNet data). On Beth Israel → Emory, keeping the

pattern was best, because there was barely any observation shift between the two (pattern-model transfer gap 0.03 AUROC, 95% CI −0.01 to 0.07); dropping it cost 0.05 AUROC. Across 26 unit pairs the pattern's meaning shifted clearly in 7, mostly for mortality; there, a single weight tuned on a 3–24 month run-in gained up to 0.02–0.03 AUROC and stayed within 0.001 elsewhere (after removing impossible lab values). A one-parameter update beat a full refit up to about 20 cases; the refit won from about 50.

3 Plan of experiments

Model. A neural latent state-space model, extending our pilot: a discrete latent patient state ($K \approx 32$ states) evolves by a learned transition, and three heads read it: a value head (likelihood of each measured value), a testing head (probability that each test is ordered, given the state) and an outcome head. Predictions come from filtering the state over time, trained end to end on the outcome loss plus the likelihood of values and tests. The testing pattern enters only through the testing head's log-likelihood, which is multiplied by the weight(s) of rungs 1–2; rung 3 re-learns the testing head with transition, value and outcome heads frozen; rung 4 fine-tunes everything. In simulation, this model trained at A and tempered by w followed the exact optimal trade-off between A and B to within 0.001 Brier. Baselines: keep (w=1), remove (w=0, Futoma), recalibration, and a standard GRU-D (fine-tuned at the last layer or in full). Lessons from our pilots are built in: tuned weights are shrunk towards 1, calibration is fitted separately from the weights, and impossible values are removed before recalibration.

E1 Simulation with known optimum. Our latent-state simulator, whose exact Bayes predictor is known, with two shift types: global (B tests less selectively) and test-specific (one test group becomes routine). Regret of each rung against the Bayes predictor at B, for run-ins of 5–200 outcome events. Expected: rung 1 wins under global shift and small run-ins, rungs 2–3 under test-specific shift once enough events exist.

E2 Real hospitals (dataset to be determined). Requirements: several hospitals with hospital identifiers, time-stamped labs and vitals with the testing pattern preserved, and enough outcome events per site; the main candidate is eICU-CRD (208 US hospitals [23]; credentialed access). Design: ordered source → target pairs, run-ins of 1, 3, 6, 12 and 24 months drawn from the target site (20 draws each), evaluation on all remaining target patients, primary metric AUROC (pre-specified), secondary calibration.

E3 Architecture. Whether the state-space model adapts from fewer cases than GRU-D fine-tuned at the last layer or in full, on the pairs of E2.

Computational cost and feasibility. All models are small (about 32 latent states or GRU hidden size 32–64, at most 24–48 hourly steps per patient, a few thousand patients per site), so a CPU is enough; we have a 4-core laptop and can request ETH's Euler cluster if needed. Measured on the laptop: the pilot state-space model trained in 15 minutes; our simulation grid of 108 deep fits (GRU, GRU-D, Neural CDE, Transformer) took 3.6 hours.

Component	Units	CPU-hours (estimate)
E1 simulation: source fits and all rungs	2 shifts × 6 run-in sizes × 20 draws	6
E2/E3 source models (∼15 min)	8 sites × 2 models × 3 seeds	12
Rungs 1–4, 2 GRU-D tunings (∼20 s)	25 pairs × 5 sizes × 20 draws × 6	83
Total		≈100

The table assumes about 8 sites and 25 pairs; cost grows linearly with the number of pairs. About 100 CPU-hours take about one day on the laptop with four workers, or a few hours on Euler. The simulator, pilot model and run-in pipeline already exist. Timeline: weeks 1–2 the multivariate state-space model and E1; weeks 3–4 E2 and E3; weeks 5–6 analysis and report.

References

[1] Sisk R, Lin L, Sperrin M, et al. Informative presence and observation in routine health data: a review of methodology for clinical risk prediction. *JAMIA* 2021;28(1):155–166.

[2] Reyna MA, Josef CS, Jeter R, et al. Early prediction of sepsis from clinical data: the PhysioNet/Computing in Cardiology Challenge 2019. *Critical Care Medicine* 2020;48:210–217.

[3] Sendak MP, Ratliff W, Sarro D, et al. Real-world integration of a sepsis deep learning technology into routine clinical care: implementation study. *JMIR Medical Informatics* 2020;8(7):e15182.

[4] Checkley W, Martin GS, Brown SM, et al. Structure, process, and annual ICU mortality across 69 centers. *Critical Care Medicine* 2014;42:344–356.

[5] Che Z, Purushotham S, Cho K, Sontag D, Liu Y. Recurrent neural networks for multivariate time series with missing values. *Scientific Reports* 2018;8:6085.

[6] Kidger P, Morrill J, Foster J, Lyons T. Neural controlled differential equations for irregular time series. *NeurIPS* 2020.

[7] Rubanova Y, Chen RTQ, Duvenaud D. Latent ordinary differential equations for irregularly-sampled time series. *NeurIPS* 2019.

[8] Shukla SN, Marlin B. Multi-time attention networks for irregularly sampled time series. *ICLR* 2021.

[9] Zhang X, Zeman M, Tsiligkaridis T, Zitnik M. Graph-guided network for irregularly sampled multivariate time series. *ICLR* 2022.

[10] Alaa AM, Hu S, van der Schaar M. Learning from clinical judgments: semi-Markov-modulated marked Hawkes processes for risk prognosis. *ICML* 2017, PMLR 70:60–69.

[11] Jeanselme V, Martin G, Peek N, et al. DeepJoint: robust survival modelling under clinical presence shift. NeurIPS Workshop on Learning from Time Series for Health, 2022. arXiv:2205.13481.

[12] Moor M, Bennett N, Plečko D, et al. Predicting sepsis using deep learning across international sites: a retrospective development and validation study. *eClinicalMedicine* 2023;62:102124.

[13] van de Water R, Schmidt H, Elbers P, et al. Yet another ICU benchmark: a flexible multi-center framework for clinical ML. *ICLR* 2024.

[14] Futoma J, Simons M, Doshi-Velez F, Kamaleswaran R. Generalization in clinical prediction models: the blessing and curse of measurement indicator variables. *Critical Care Explorations* 2021;3(7):e0453.

[15] Rockenschaub P, Xian Z, Zamanian A, et al. Robust prediction under missingness shifts. arXiv:2406.16484, 2024.

[16] Lasko TA, Strobl EV, Stead WW. Why do probabilistic clinical models fail to transport between sites. *npj Digital Medicine* 2024;7:53.

[17] Zhou H, Balakrishnan S, Lipton Z. Domain adaptation under missingness shift. *AISTATS* 2023, PMLR 206:9577–9606.

[18] Steyerberg EW, Borsboom GJ, van Houwelingen HC, et al. Validation and updating of predictive logistic regression models: a study on sample size and shrinkage. *Statistics in Medicine* 2004;23:2567–2586.

[19] Vergouwe Y, Nieboer D, Oostenbrink R, et al. A closed testing procedure to select an appropriate method for updating prediction models. *Statistics in Medicine* 2017;36:4529–4539.

[20] Riley RD, Snell KIE, Archer L, et al. Evaluation of clinical prediction models (part 3): calculating the sample size required for an external validation study. *BMJ* 2024;384:e074821.

[21] Lee Y, Chen AS, Tajwar F, et al. Surgical fine-tuning improves adaptation to distribution shifts. *ICLR* 2023.

[22] Kirichenko P, Izmailov P, Wilson AG. Last layer re-training is sufficient for robustness to spurious correlations. *ICLR* 2023.

[23] Pollard TJ, Johnson AEW, Raffa JD, et al. The eICU Collaborative Research Database, a freely available multi-center database for critical care research. *Scientific Data* 2018;5:180178.